

YA'AN, China

WMO: 562870

Lat: 29.9830N

Lon: 102.9966E

Elev: 629

StdP: 93.99

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.4	2.5	-2.5	3.3	5.2	-1.0	3.7	5.2	4.5	7.8	4.1	7.9	0.8	40	0.308

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.9	33.4	24.9	32.1	24.4	30.9	23.7	26.2	31.5	25.4	30.4	24.7	29.2	1.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	21.3	29.4	24.1	20.5	28.3	23.5	19.8	27.5	86.1	31.7	82.6	30.5	79.5	29.3	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.5	4.0	3.4	-0.1	35.2	1.2	1.1	-0.9	36.0	-1.6	36.7	-2.3	37.3	-3.2	38.0		
			-0.8	27.4	1.4	0.8	-1.8	27.9	-2.7	28.4	-3.4	28.8	-4.5	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.3	6.8	8.9	13.2	18.2	21.7	24.2	26.3	26.0	22.0	17.7	13.2	8.3
	DBStd	7.30	2.50	3.09	3.68	3.68	2.77	2.51	2.29	2.61	2.74	2.62	2.93	2.18
	HDD10.0	234	103	53	11	1	0	0	0	0	0	0	6	61
	HDD18.3	1345	358	264	162	46	4	0	0	0	2	43	154	311
	CDD10.0	2886	4	22	111	248	363	425	504	497	361	239	103	8
	CDD18.3	954	0	0	4	43	109	175	246	239	113	24	1	0
	CDH23.3	6524	0	0	22	232	588	1131	2049	1957	494	50	0	0
	CDH26.7	2127	0	0	2	51	145	346	737	722	121	5	0	0
Wind	WSAvg	1.3	1.0	1.1	1.3	1.4	1.6	1.5	1.5	1.5	1.2	1.0	0.9	1.0
Precipitation	PrecAvg	1724	22	31	52	84	131	177	370	445	214	108	58	22
	PrecMax	2368	63	72	137	189	310	410	758	809	618	176	144	39
	PrecMin	1207	3	11	15	27	45	94	147	118	75	49	24	8
	PrecStd	310	12	14	25	39	55	79	152	177	117	30	27	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.1	19.8	25.8	30.2	31.6	33.2	34.9	35.0	32.3	27.1	22.1	15.6
		MCWB	9.4	12.2	16.3	19.4	20.9	23.1	25.2	25.8	24.4	19.8	16.1	10.3
	2%	DB	13.2	17.2	23.0	28.1	29.9	31.6	33.4	33.5	29.7	24.7	20.2	13.9
		MCWB	8.2	11.2	15.1	18.8	20.2	22.7	25.2	25.3	23.1	19.0	15.4	9.4
	5%	DB	11.5	15.2	20.8	26.1	28.3	30.2	32.1	32.1	27.9	22.9	18.4	12.5
		MCWB	7.6	10.3	14.3	18.2	19.9	22.3	24.8	25.0	22.1	18.5	14.8	9.1
	10%	DB	10.1	13.4	18.7	24.0	26.7	28.7	30.7	30.7	26.1	21.2	17.0	11.3
		MCWB	7.1	9.7	13.4	17.5	19.4	22.0	24.4	24.5	21.4	17.9	14.2	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.1	12.7	18.1	21.3	22.8	25.2	27.2	27.2	25.9	21.4	17.0	11.6
		MCDB	14.8	18.1	23.7	27.7	28.3	30.0	32.8	32.8	30.5	24.8	20.4	13.9
	2%	WB	8.9	11.7	16.0	19.9	21.7	24.2	26.2	26.2	24.3	19.9	16.1	10.6
		MCDB	11.9	15.9	20.8	25.5	27.1	28.9	31.7	31.5	28.0	23.1	19.1	12.4
	5%	WB	8.1	10.9	14.9	18.8	20.8	23.5	25.5	25.5	23.1	19.1	15.3	9.7
		MCDB	10.8	14.3	19.6	24.3	26.1	28.0	30.5	30.5	26.4	21.7	17.6	11.6
	10%	WB	7.4	10.0	14.0	17.9	20.1	22.7	24.8	24.8	22.1	18.3	14.7	9.0
		MCDB	9.8	13.0	18.1	23.0	25.0	26.9	29.2	29.5	25.2	20.8	16.5	10.9
Mean Daily Temperature Range	5% DB	MDBR	4.5	5.2	6.7	7.6	7.3	6.9	6.9	6.9	5.5	4.9	4.5	4.4
		MCDBR	7.5	8.9	10.3	10.8	10.1	9.5	9.4	9.1	8.5	7.6	7.0	6.9
		MCWBR	3.8	4.3	4.5	4.3	4.1	4.0	4.1	3.8	3.5	3.2	3.2	3.6
	5% WB	MCDBR	6.6	8.0	9.1	9.6	8.5	8.0	8.4	8.3	7.3	6.4	5.8	5.5
		MCWBR	3.5	3.9	4.2	4.2	3.9	3.7	3.9	3.7	3.2	2.9	2.9	3.0
Clear-Sky Solar Irradiance	taub	0.622	0.658	0.714	0.707	0.676	0.636	0.672	0.700	0.698	0.683	0.660	0.623	
	taud	1.577	1.543	1.493	1.553	1.651	1.768	1.695	1.632	1.615	1.597	1.572	1.597	
	Ebn at Noon	581	612	618	648	673	699	671	646	622	587	547	552	
	Edh at Noon	229	257	288	279	254	225	241	255	251	240	228	215	
All-Sky Solar Radiation	RadAvg	2.33	2.79	3.32	3.97	4.05	3.77	4.07	3.90	2.94	2.43	2.30	2.13	
	RadStd	0.28	0.30	0.31	0.37	0.43	0.49	0.51	0.58	0.28	0.31	0.16	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (5 neighbors)	+0.31	N/A	N/A	+0.49	N/A	N/A	-50	-91	+164	+97

Nomenclature: See separate page